

Discrete Math Cheat Sheet

Logic (Propositions & Predicates)

- Example: 'If it rains, then I stay home.'
- Propositional logic uses symbols like p , q .
- Predicate logic adds variables: $P(x)$: 'x is raining'.

Relations and Functions

- Set: $A = \{1, 2, 3\}$
- Sequence: 1, 2, 3, 4...
- Relation: 'is friend of'
- Equivalence Relation: reflexive, symmetric, transitive.
- Function: $f(x) = x + 1$.

Proofs and Techniques

- Direct Proof: Show 'if p then q '.
- Contrapositive Proof: Show 'if not q then not p '.
- Proof by Contradiction: Assume not p , find a contradiction.

Counting (Combinatorics)

- Addition Principle: 3 ways + 2 ways = 5 ways.
- Multiplication Principle: 3 shirts x 2 pants = 6 outfits.
- Permutations: Order matters ($n!$).
- Combinations: Order doesn't (nCr).